CHAPTER THREE

Reduced Subsurface Lateral Flow in Agroforestry System Is Balanced by Increased Water Retention Capacity: Rainfall Simulation and Model Validation

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Contents

Abstract

Soil hydrology controls the terrestrial water cycle and the transport of substances to influence the environmental quality of Earth's critical zone (CZ). Soil and water management in agroforestry systems (AF) is able to reduce soil nitrogen losses and to alleviate secondary salinity in some regions of the world by reducing subsurface lateral flow. Compared to monocropping (MC) system, the reduction of subsurface lateral flow in AF has been attributed not only to the enhanced evapotranspiration and canopy

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interception but also to changes in soil structure and related hydraulic properties. However, for AF, it remains unclear how changes in soil structure and hydraulic properties occur and can act to reduce the subsurface later flow. Rainfall simulation experiments were conducted in the field and soil matric potential was measured to determine the effect of AF and MC on the dynamics of rainfall infiltration, subsurface lateral flow, and soil water storage in the soil profile. The calculated isolines of soil matric potential showed that a physical domain of water saturation occurred in the subsoils during the rainfall and diminished after the rainfall. The water saturation domain was larger during the rainfall and drained more slowly after the rainfall in AF than in MC. These results illustrated that AF increased vertical preferential flow and retarded the subsurface lateral flow, resulting in increased water retention capacity in the soil profile, compared to MC. The changed water mass and flow distribution was attributed to the deep roots, which increase macropores oriented in the vertical direction and modify micro- and mesopores in the lateral direction, resulting in changes in anisotropy of soil hydraulic properties along transects of slope. These proposed mechanisms were successfully verified by mathematical modeling. Numerical experiments using the Hydrus-2D mathematical modeling code at the virtual condition of the same antecedent soil moisture condition along the slope at different rainfall events ruled out the effect of antecedent soil moisture or evapotranspiration on generation of subsurface flow. These findings suggest that land use has strong effects on water distribution not only above the ground but also in the subsurface. The changes in soil structure and hydraulic properties need to be considered in understanding landscape hydrology related to agricultural practices and their impacts on Earth's CZ.

1. INTRODUCTION

Soil is the central interface at the heart of Earth's critical zone (CZ), representing a natural geomembrane across which water is actively exchanged with the atmosphere, biosphere, hydrosphere, and lithosphere (Brantley et al., 2007; Lin, 2010). Subsurface lateral flow is generally induced when sufficient infiltrating water is impeded over a less permeable layer or bedrock within soil profiles (Wang et al., 2011). The resulting changes in the water flow field significantly contribute to the transport and transformation of energy and material within the CZ (Lin, 2010; Schaik, 2010; Wang et al., 2011). As subsurface lateral flow occurs within soil, the governing processes are ruled by soil properties (Lin, 2010; Schaik, 2010). Land use, evolving with different natural and anthropogenic processes, has strong influences on soil structure and hydraulic properties (Bottinelli et al., 2016; Schaik, 2010; Zhang et al., 2015a,b). Once a small-scale change in soil structure and hydraulic properties occurs, a relatively large-scale variation in subsurface lateral flow can be induced (Schaik, 2010; Zhang et al., 2015a,b). Although numerous studies have intensively investigated subsurface lateral flow under various land uses (Gerrits, 2010; Wang et al., 2011), few have explored the relationship between land use and subsurface lateral flow from the view of a controlling role for soil structure and hydraulic properties.

Soil structure is the physical architecture that arises from the physical arrangement of the pore spaces among closely packed soil particles, including minerals, decaying biomass, and living organisms. There are many possible natural and anthropogenic factors that influence soil structure and hydraulic properties under various land uses. Vegetation species significantly influence the development of soil structure through the inputs of below-ground biomass and exudates (Nair et al., 2010), which influences the microbiome and aggregation of soil particles and alters the pore space architecture. Through this, vegetation species influence the soil hydraulic conductivity and the soil water retention capacity (Nath, 2015; Seobi et al., 2005; Wang et al., 2013). Plant root penetration and tillage may create soil macropores with low water entry pressures, high connectivity, and large cross section which accelerates soil water flow along the orientation of the macropore (Bogner et al., 2010; Schaik, 2010; Zhang et al., 2015a,b). Plant root shape may result in significant anisotropy in soil hydraulic properties, reflecting an increase in the spatial variability in vertical preferential flow and subsurface lateral flow (Jing et al., 2008; Schaik, 2010). The impacts of land use change on soil structure and hydraulic properties can occur rapidly, within a single season, in natural and agricultural ecosystems (Schaik, 2010; Seobi et al., 2005). These changes can have immediate influences on subsurface lateral flow process (Bogner et al., 2010; Schaik, 2010), although quantitative effects on the flow field are not understood.

Agroforestry system (AF) can reduce subsurface lateral flow in the downward direction along slopes (Dunin, 2002; Wang et al., 2011). The mechanisms behind this effect have been intensively explored with respect to atmospheric and biospheric processes, i.e., the characteristics of precipitation, evapotranspiration, and canopy interception (Dunin, 2002; Ticehurst et al., 2001; Wang et al., 2011), but rarely with consideration of soil hydrological properties and processes. Mapa (1995) has observed the greatest soil water retention at any given pore entry pressure in reforested soil profiles, in comparison with the observations on cultivated and grassland soils. Other studies also suggest that tree roots are able to create well-connected macropores or channels to accelerate soil water flow (Bogner et al., 2010). Recently, some studies suggest that tree-based ecosystems increase soil carbon sequestration in deeper soil due to greater root density and larger carbon inputs compared to tree-free ecosystems (Nair et al., 2010), which thereby leads to changes in soil porosity and associated soil hydraulic properties in tree-based ecosystems (Nath, 2015). Zhang et al. (2015a,b) also reported that a high tree root density had a pivotal influence on preferential flow in forest soils. These findings imply that tree-based agroforestry may strongly influence subsurface lateral flow by altering soil structure and hydraulic properties.

Numerous studies have attempted to explore the mechanisms of subsurface lateral flow using virtual modeling, or numerical experiments with a model that is driven by collective field intelligence (Gerrits, 2010; Keim et al., 2006; Wang et al., 2011). For example, Keim et al. (2006) applied the Hydrus-2D modeling code to assess the impacts of tree canopy on subsurface lateral flow by using measured rainfall and synthetic and smoothed throughfall. Gerrits (2010) separated the impacts of interception between tree canopy and forest floor on subsurface lateral flow by simulating the conditions of no interception, canopy interception only, and both canopy and forest floor interception. However, virtual modeling often lacks field validation, and hence its application does not fully develop insight into the fundamental physical mechanisms of subsurface flow. Experimental rainfall simulation is commonly used for furthering understanding on subsurface lateral flow process, because it allows predefining some key parameters like rainfall duration and quantity (Wildhaber et al., 2012). Examples of this approach include recent work by Zhao et al. (2014) and Fu et al. (2015). In the study reported here, a combination of field rainfall simulation experimentation and numerical modeling is applied together to determine the mechanism of reduced subsurface lateral flow in an AF in compared to a monocropping (MC) system. The AF consisting of citrus (*Citrus reticulata*) trees with intercropped peanut (*Arachis hypogaea*) was compared with an adjacent MC system of peanut. The objectives of this study were (i) to characterize the mechanistic connections between vertical preferential flow, subsurface lateral flow, and plant root systems and (2) to explore using numerical modeling the mechanisms through which AF can reduce subsurface lateral flow. The numerical modeling with the Hydrus-2D code was developed using field-collected soil hydraulic properties, slope structure, and rainfall intensities and virtually designed antecedent soil matric potentials. The hypothesis of this study is that changes in soil structure and hydraulic properties between AF and MC lead to variation in subsurface lateral flow in the downward direction along slopes.

2. MATERIALS AND METHODS

2.1 Study Site and Land Uses

The field experiment was carried out in a small agricultural catchment (Tang et al., 2011; Zepp et al., 2005), about 5 km from the Ecological Experimental Station of Red Soil, Chinese Academy of Sciences (EESRS, CAS) in Yingtan, Jiangxi province, China (28°15′N, 116°55′E). The elevation at the catchment is 41–55 m above mean sea level. The study area has a subtropical warm humid climate, with an annual mean air temperature of 17.8°C, a minimum of 5.3°C in January, a maximum of 29.6°C in July, and a frost-free period of 262 days. The annual rainfall average is 1706 mm during from 1954 to 1999, with approximately 50% concentrated during the period from April to July. The hillslope gradient was 6% and the slope length was 130–150 m. The soil depth along the slope ranged from 0.4 to 2.0 m overlying an impermeable parent material horizon. The soils developed from Quaternary clay are classified as a loam clay Ultisol according to the USDA soil taxonomy (Soil Survey Staff, 2010).

The two agroecosystems were established after a clearance of tea plants (*Camellia sinensis* L.) in 1982. The two systems were established adjacently along the downward trajectory of the hillslope. The citrus trees were spaced at $4 \times 4\ m^2$ along the slope and peanut crops were sown in a 1.5 m wide alley between the tree rows. For this study, the trees were approximately 3.0 m high, with a canopy width of 3–4 m. The root depth was about 1.1 m (Wang et al., 2011). The peanut crops were sown horizontally along the slope at a spacing of $0.2 \times 0.3\ m^2$ in rows in both systems. Peanut crops grew from the middle of April to the middle of August and were managed in the same way in both the agroecosystems.

Soil pits were dug in both systems and characterized following soil survey protocols (FAO/SRIC/ISSS, 1998). The tree root density in a $0.5 \times 0.5\ m^2$ area was counted and soil samples were taken for physical analysis at each soil horizon. Soil particle size distribution was measured with the pipette method (Lu, 1999). Five bulk soil cores were taken at each soil horizon, and then air-dried and weighed in the laboratory to determine bulk soil density. In AF, four horizontal frames of $1.0 \times 0.5\ m^2$ in each soil horizon were exposed vertically, sequentially during pit digging, and the roots in each horizontal frame were counted. The selected soil properties and root distribution are presented in Table 1.

Table 1 Selected Soil Physical Properties at Different Depths in the Monocropping (MC) and Agroforestry (AF) Systems

Land Use	Soil Horizon[a]	Depth (m)	Bulk Density (Mg m^{-3})	Particle Size Distribution (%)			Comments[b]
				0.05–2.0 mm	0.002–0.05 mm	<0.002 mm	
Monocropping (MC)	A_p	0–0.2	1.36	42.49	24.65	32.86	Prismatic/blocky; rich in biopores formed from decayed root
	AB	0.2–0.5	1.43	38.19	25.96	35.85	Prismatic/blocky; some biopores formed from decayed root
	B_t	0.5–0.9	1.51	36.45	25.75	37.80	Granular/crumbs
	BC_v	0.9–1.5	1.60	34.04	26.79	39.17	Coherent and single grain
Agroforestry (AF)	A_p	0–0.2	1.35	41.81	25.80	32.39	Prismatic/blocky; tree root density of 1522 m^{-2}
	AB	0.2–0.5	1.47	39.56	26.18	34.26	Prismatic/blocky, tree root density of 725 m^{-2}
	B_t	0.5–0.9	1.51	35.69	26.71	37.60	Granular/crumbs; tree root density of 307 m^{-2}
	BC_v	0.9–1.5	1.60	32.07	26.93	41.00	Coherent and single grain; tree root density of 175 m^{-2}

[a]The small letters for soil horizons are: t, accumulation of silicate clay and v, plinthite.
[b]Root density, number of roots per square meter.

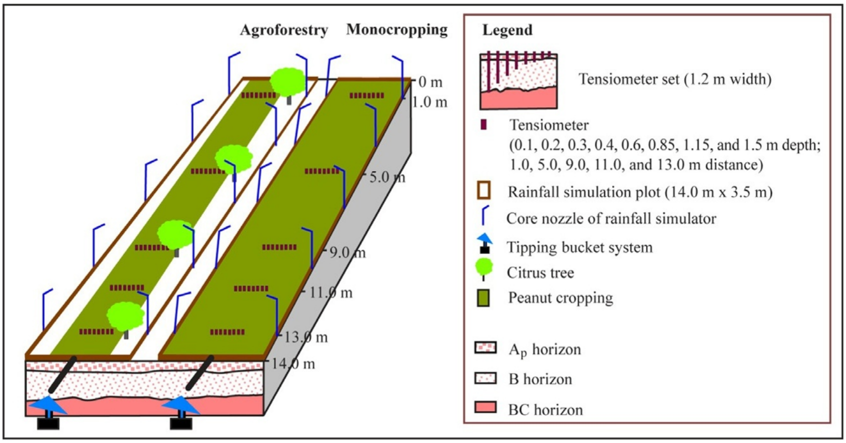

Fig. 1 The equipment layout from rainfall simulation experiment in the erosion plots under the monocropping and agroforestry systems.

2.2 Rainfall Simulation

The rainfall simulation was conducted by using rainfall sprinklers in two erosion plots located at the same slope position under both land uses (Fig. 1). The erosion plots enclosed with metal plates were 3.5 m wide and 14.0 m long along the slope. The metal plates were inserted into 0.2 m soil depth and projected 0.2 m above the ground. The sprinklers made in Germany, consisted of 10 core spraying nozzles that were positioned at 1.0, 4.0, 7.0, 10.0, and 13.0 m downslope from the upper boundary of each plot and mounted 3.0 m above the ground. The rainfall simulations were performed during periods without wind; water was supplied to control rainfall intensity by changing the number of nozzles. At the lower end of each erosion plot, one tipping bucket system was installed to collect overland flow during rainfall simulation and the number of bucket tips were recorded using an event data logger (Onset Computer Corporation, USA), then calibrated for flow rate using the total overland flow volume that was collected. Rainfall simulation experiments were performed over 2 h from 17:00 to 19:00 on six different days during the period 1st–18th April, 2008 (Eq. 1). Before rainfall simulation, the soil was tilled with moldboard plow and peanuts were not sown. Each simulated rainfall event was applied for 2 h to both plots at the same time. During rainfall simulation, rainfall volume was measured by using eight cups randomly distributed in the peanut cropping area and another eight cups placed at different positions underneath the tree canopy. Water from the cups was weighed to determine rainfall in both plots and canopy throughfall in the AF. The six rainfall simulations were initiated when the soil matric potential was approximately −80 hPa at the 0.1 m depth and −0 hPa at the 1.50 m depth.

Two weeks before the rainfall simulation, five sets of tensiometers were installed at the 1.0, 5.0, 9.0, 11.0, and 13.0 m distance downslope from the upper boundary of each plot to determine the spatial and temporal variations of soil matric potential. The sets of tensiometers were 0.1, 0.2, 0.3, 0.4, 0.6, 0.85, 1.15, and 1.5 m deep and installed at a width of 1.2 m (Fig. 1). Therefore, a grid of 40 tensiometers was installed in each plot. Readings of the soil matric potential were recorded at 10-min intervals using a data logger (DL2e, Delta T Inc., UK) connected with pressure transducers installed in the tensiometers. The soil matric potential was negative in the unsaturated zone, positive in the saturated zone, and zero at the interface. Soil water storage was demonstrated by the difference and the average of soil matric potential before and after each simulated rainfall in the soil profiles for both agroecosystems. For the rainfall events of 95.6 mm that was equal in both plots (see in Table 2), catenary variations of soil moisture were presented for 10 h continuously from the beginning of each 2-h simulated rainfall. These variations were represented by isolines of soil matric potential based on conventional Kriging interpolation on a rectangular grid using horizontal and vertical georeference points.

Table 2 Estimated Saturated Soil Hydraulic Conductivities (K_s) and van Genuchten Parameters ($\theta(\psi_m)$) at Different Soil Depths in the Monocropping (MC) and Agroforestry Systems (AF) Through Hydrus-2D Inversion Modeling for the Rainfall Events of 95.6 mm

			van Genuchten Parameters of $\theta(\psi_m)$[a]			
Land Use	**Depth (m)**	**K_s[a] (m h^{-1})**	**θ_r (m^3 m^{-3})**	**θ_s[a] (m^3 m^{-3})**	**a (m^{-1})**	**n**
Monocropping (MC)	0–0.2	10	0.33	0.50	14.5	2.68
	0.2–0.9	80	0.30	0.44	80	1.33
	0.9–1.5	18	0.41	0.43	34	2.68
Agroforestry (AF)	0–0.2	10	0.30	0.50	19.5	2.68
	0.9–0.5	80	0.25	0.45	68	1.38
	0.9–1.5	13	0.31	0.44	34	2.66

[a]K_s, saturated hydraulic conductivity; $\theta(\psi_m)$, water retention curve; θ_s, saturated soil water content were measured by Jing et al. (2008).

2.3 Estimation of Subsurface Lateral Flow by Water Balance

Assuming that the horizon of parent material below the soil profiles was impermeable, such that the infiltrating water would produce subsurface lateral flow along the slope, the subsurface lateral flow (Q_{inter}, mm) was quantified by the water mass balance (Eq. 1).

$$Q_{inter} = P - P_{int} - R - ET - \Delta S \tag{1}$$

where P is total precipitation (mm), calibrated using plastic cups in the field; P_{int} is canopy interception (mm) measured in AF; R is overland flow (mm), measured using the tipping bucket system; ET is evapotranspiration (mm), which was assumed zero as the rainfall simulation was conducted in the evening under humid air and low temperature; and ΔS is change in soil water storage (mm), calculated based on measured soil moisture in the soil profile and the soil water retention curve reported for the same slope position by Jing et al. (2008).

2.4 Inverse Modeling of Subsurface Lateral Flow

Saturated soil hydraulic conductivities (K_s) and van Genuchten parameters ($\theta(\psi_m)$) in each agroecosystem were estimated through inverse modeling capabilities of the Hydrus-2D code (Simunek et al., 1996), based on the soil matric potentials measured during the rainfall events. To reduce the uncertainty in the estimated parameter values, the saturated soil water contents (θ_s) were fixed to the observed field values (Jing et al., 2008). The dimensions of the calculation domain were set to the size of the rectangular plots, being 14.0 m long and 2.0 m deep (Fig. 2). The slope gradient was 6%. The upper

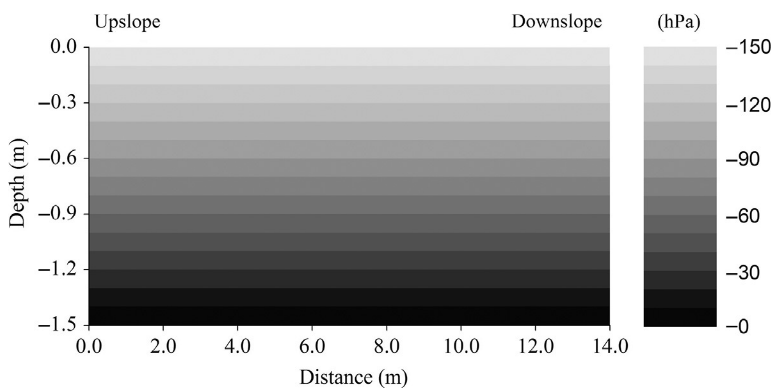

Fig. 2 The slope domain for Hydrus-2D modeling, with the 6% slope gradient georeferenced into rectangular. The isolines of soil matric potential in the soil profile illustrate the equilibrium along the slope, set up for the virtual modeling.

boundary was the atmospheric boundary. The bottom boundary was defined as zero flux due to the impermeable horizon of the parent material. The upper vertical boundary was defined as a no-flux boundary and the lower vertical boundary was defined as a free drainage boundary. The horizons of the soil profiles, i.e., layers of depths 0–0.2, 0.2–0.9, and 0.9–1.5 m, were defined as homogenous in terms of soil physical and hydraulic properties. Soil matric potentials, measured at the 0.1, 0.4, and 1.15 m depth at the 13.0 m distance from the upslope vertical boundary, were chosen as the soil horizons to perform the inverse modeling. To avoid possible numerical instability in the case of saturation water excess as overland flow, actual infiltration water (I, mm) (Eq. 2) was used as the input variable instead of rainfall during modeling. The actual infiltration was calculated with Eq. (2):

$$I = P - P_{int} - R \tag{2}$$

where P is total precipitation (mm), calibrated using plastic cups in the field; P_{int} is canopy interception (mm) measured in the AF plot; and R is overland flow (mm), measured using the tipping bucket system.

The goodness of fit was evaluated by comparing the simulated and measured soil matric potentials for the simulated rainfall events. The estimated values for saturated soil hydraulic conductivities and van Genuchten parameters were further validated by comparing the modeled and the calculated subsurface lateral flow. These comparisons for goodness of fit were carried out for all rainfall events. The coefficient of determination (R^2) and relative error (RE, %) were calculated to quantify the goodness of fit and the validity of the estimated parameter as follows (Eqs. 3 and 4):

$$R^2 = 1 - \frac{\sum_i (M_i - \overline{M_i})^2}{\sum_i (M_i - S_i)^2} \tag{3}$$

$$RE = \frac{M_i - S_i}{M_i} \times 100\% \tag{4}$$

where M_i is measured soil matric potential (hPa) or calculated subsurface lateral flow (mm); S_i is modeled soil matric potential or subsurface lateral flow simulated by Hudrus-2D; and $\overline{M_i}$ is the mean value of M_i.

The antecedent soil matric potential showed strong spatial heterogeneity (Fig. 3). This heterogeneity could create uncertainty in determining the effects of soil structure and hydraulic properties on subsurface lateral flow process along hillslopes. Therefore, a numerical experiment with Hydrus-2D was

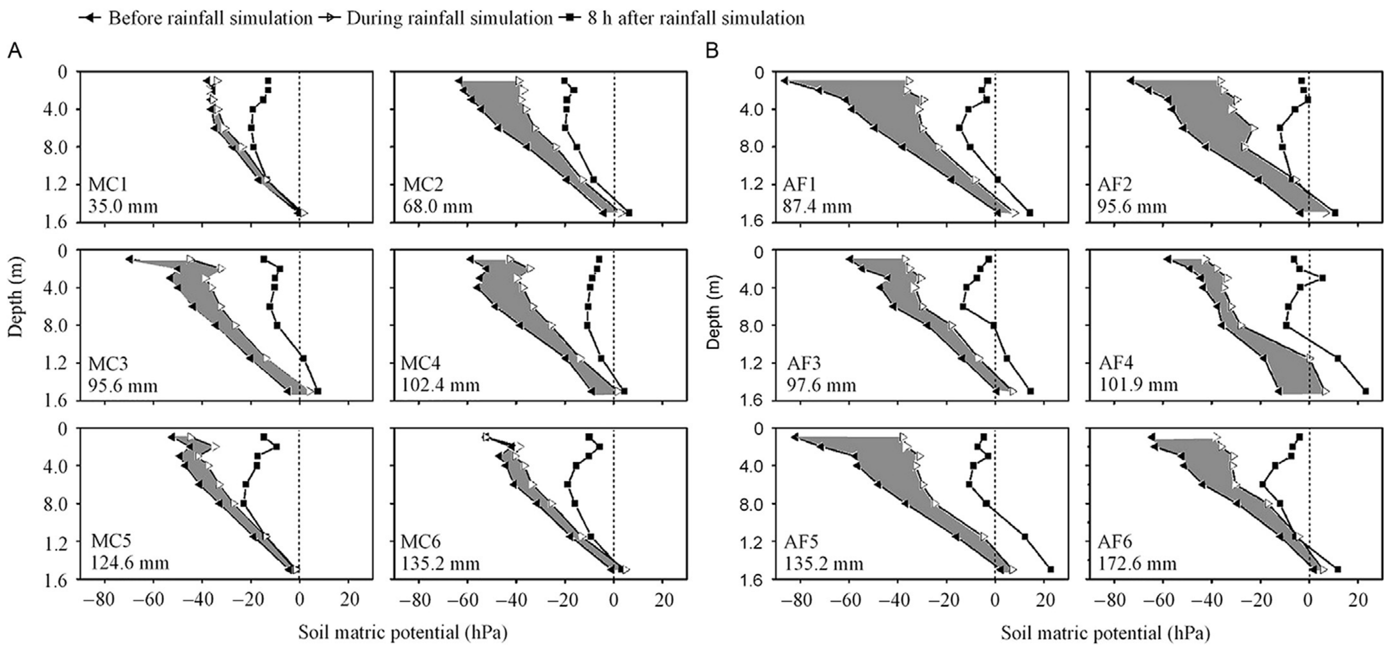

Fig. 3 Average soil matric potential before, during, and after rainfall simulation in the (A) monocropping (MC) system and (B) agroforestry (AF) system. The *shadow* indicates variation of soil matric potential between before and after rainfall simulation. The numbers following MC and AF were the sequence numbers of simulation rainfall events listed in Table 2.

performed under the virtual condition of the same antecedent soil matric potential along the slope for all simulated rainfall events to determine the effect of land use on subsurface lateral flow, the changes in soil water storage, and temporary water table elevation. The virtual soil matric potential profile was set up as equal distribution from −150 to 0 hPa at intervals of 10 cm in the 1.5-m depth soil profile (Fig. 2). When the two land uses have similar responses to different rainfall intensities and the effects of antecedent soil moisture can be ruled out, the results are designed to shed light on mechanisms of reduced subsurface lateral flow in AF systems, in comparison to MC through modification of soil structure and hydraulic properties.

3. RESULTS

The profiles of soil matric potential are plotted for all six simulated rainfall events (Fig. 3). The increase in soil matric potential before and 8 h after rainfall simulation represents the variation in soil water storage. After the rainfall simulation, the increased magnitude of the water storage in the soil profiles was larger for the agroforestry (AF) treatment than for the MC treatment in all six rainfall events. The soil matric potential in the lower soil layer 8 h after rainfall was near zero in MC and more positive, or demonstrating evidence of a higher temporary water table, in AF. Moreover, the soil matric potential during rainfall showed a different distribution in the upper soil profiles between the two agroecosystems, and it was greater at the 0.3–0.4 m depth than adjacent soil depths in AF, deeper than at the 0.2 m depth in MC.

The catenary variation of soil matric potential over time for the rainfall events of 95.6 mm demonstrates the effects of agroecosystems on subsurface water distribution (Fig. 4). The isolines of soil matric potential show that antecedent soil moisture before the rainfall (0 h) was similarly evenly distributed along the slope in both agroecosystems. There was a slightly drier condition at the downslope location in AF, compared with this location in MC. The isoline of zero soil matric potential shows that water saturation domains occurred not only in the surface soil but also in the deepest soil layers studied. At 0.5 h after the beginning of the rainfall event, the domains with water saturation were disconnected between the surface soil and the deep soil and between the upslope and the downslope locations. At 1.0 and 1.5 h after the beginning of rainfall, the water saturation domains expanded in the deep soil and became connected between the upslope

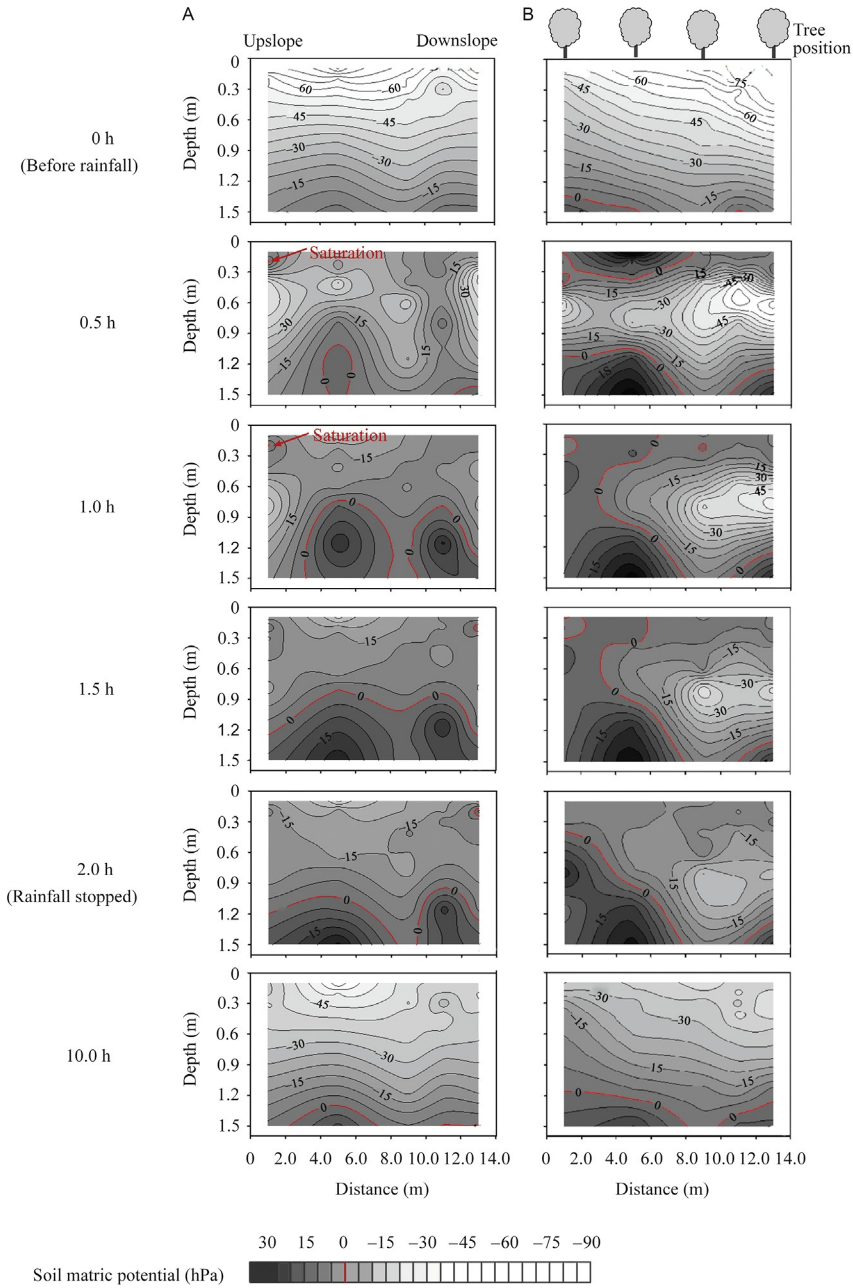

Fig. 4 Ordinary Kriging interpolation of soil matric potential (hPa) measured at different depths along slope for the simulated rainfall events of 95.6 mm in the monocropping system (A) and agroforestry system (B). The *red lines* show zero soil matric potential.

and downslope locations for the MC treatment. For the AF treatment, the water saturation domain expanded at the upslope position and became connected between the surface soil and the deep soil. For both agroecosystems, the water saturation domain became smaller following the end of the rainfall (after 2.0 h) compared to that at 1.5 h.

The modeled and measured soil matric potentials agreed well for the rainfall events of 95.6 mm (Fig. 5). However, the modeled values showed somewhat of a delay compared to the measured responses at the beginning of the rainfall, particularly at the 0.4 and 1.15 m depths in AF and at the 1.15 m depths in MC. The mean hourly determination coefficients (0.78–0.79) and RE values (14.6–15.7%) between the modeled and measured soil matric potentials were comparable in the two agroecosystems

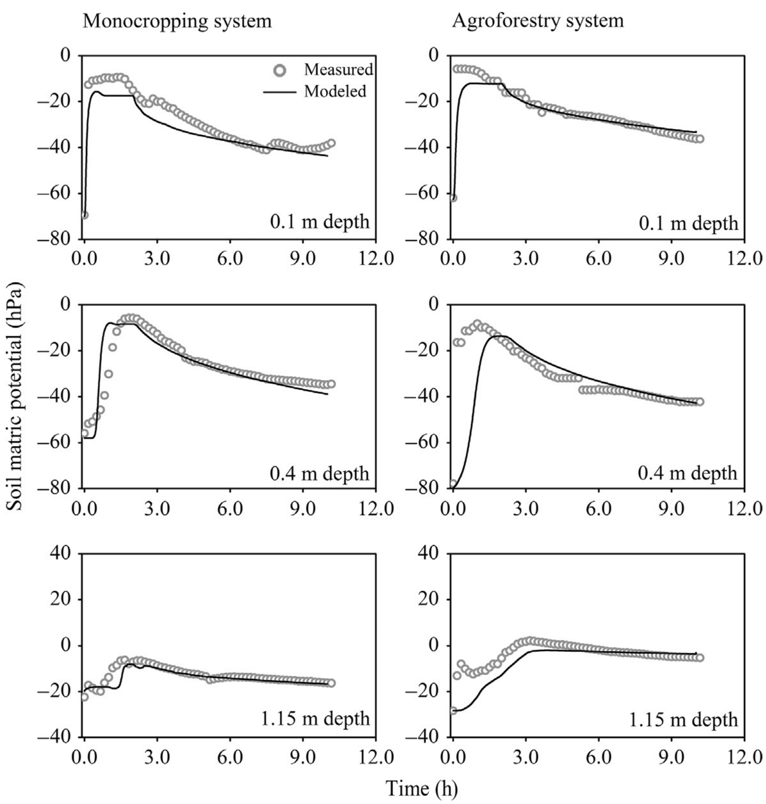

Fig. 5 Measured and modeled soil matric potentials using Hydrus-2D inversion modeling for the simulated rainfall events of 95.6 mm in the monocropping system and agroforestry system.

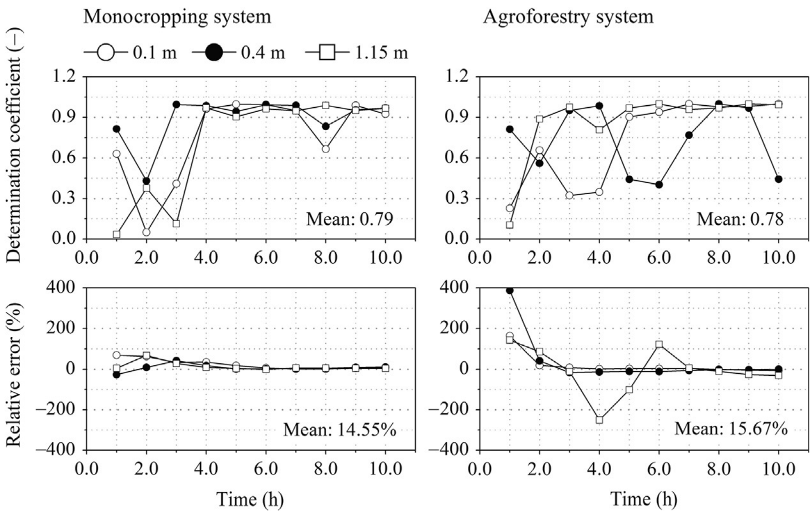

Fig. 6 Determination coefficient (R^2) and RE of soil matric potential at different soil depths between the values measured and modeled using Hydrus-2D for the simulated rainfall events of 95.6 mm in the monocropping system (MC) and agroforestry system (AF). The R^2 and RE was hourly calculated using Eqs. (3) and (4).

(Fig. 6). The estimated values for the parameters of the van Genuchten water retention curve were similar for both agroecosystems (Table 2). There was a smaller value for α at the 0–0.2 m depth (14.5 vs 19.5 m^{-1}) and at the 0.2–0.9 m depth (80 vs 68 m^{-1}), respectively, in MC compared to AF. By using estimated soil hydraulic parameters, soil water matric potentials were simulated for all rainfall events and the water mass balance was calculated (Table 3). The contributions of the water balance demonstrated that the overland flow accounted for approximately 50% the total rainfall in both agroecosystems. The infiltration water was about 16% less in AF than in MC due to approximately 12% greater canopy interception in AF. On average, the soil water storage and subsurface lateral flow accounted, respectively, for 38.7% and 61.3% of the infiltration water in AF and 11.8% and 88.2% the infiltration water in MC. The measured and simulated subsurface lateral flow was well correlated for all the rainfall events, showing high goodness of fit as indicated by the high values of determination coefficient (R^2) and low RE (Figs. 6 and 7).

Under the virtual condition of even distribution of antecedent soil matric potential, the numerical modeling results show that the two systems exhibited similar responses under different rainfall intensities in terms of predicted subsurface lateral flow and changes in soil water storage and

Table 3 Components of Water Budget for All Simulated Rainfall Events in the Monocropping (MC) and Agroforestry (AF) Systems

			Interception[b] P_{int}		Overland Flow[b] R		Infiltration I		Soil Water Storage[b] ΔS		Subsurface Lateral Flow, Q_{inter}	
Land Use	**Event[a]**	**Total Rainfall[b] P (mm)**	**mm**	**(E_{int}/P) %**	**mm**	**(R/P) %**	**mm**	**(I/P) %**	**mm**	**($\Delta S/I$) %**	**mm**	**(Q_{inter}/I) %**
Monocropping (MC)	MC1	35.0	–	–	12.3	35.1	22.7	64.9	0.9	4.0	21.8	96.0
	MC2	68.0	–	–	29.7	43.7	38.3	56.3	10.5	27.4	27.8	72.6
	MC3	95.6	–	–	52.6	55.0	43.0	45.0	4.4	10.2	38.6	89.8
	MC4	102.4	–	–	52.6	51.4	49.8	48.6	8.4	16.9	41.4	83.1
	MC5	124.6	–	–	61.3	49.2	63.3	50.8	5.5	8.7	57.8	91.3
	MC6	135.2	–	–	73.7	54.5	61.5	45.5	2.2	3.6	59.3	96.4
Agroforestry (AF)	AF1	95.6	12.5	13.1	50.9	53.2	32.2	33.7	15.8	49.1	16.4	50.9
	AF2	97.6	11.7	12.0	46.1	47.2	39.8	40.8	9.8	24.6	30.0	75.4
	AF3	87.4	14.4	16.5	51.9	59.4	21.1	24.1	16.9	80.1	4.2	19.9
	AF4	101.9	15.9	15.5	54.1	48.9	31.9	31.3	11.2	35.1	20.7	64.9
	AF5	135.2	14.0	10.4	66.2	54.7	55.0	40.7	15.3	27.8	39.7	72.2
	AF6	172.6	12.1	7.0	82.8	48.0	77.7	45.0	12.1	15.6	65.6	84.4

[a]The number following MC and AF indicates the sequence number of simulation rainfall events.
[b]Overland flow, determined by tipping bucket system; rainfall and interception, determined by plastic monitoring cups; evapotranspiration, defined as zero here associated with the humid air moisture and low temperature during rainfall simulation period; and soil water storage, calculated based on the measured soil matric potentials during the simulation rainfall and the van Genuchten parameters determined by Jing et al. (2008).

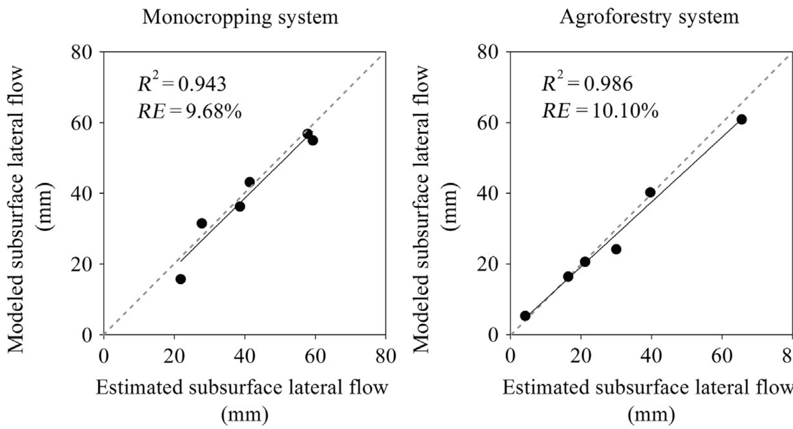

Fig. 7 The modeled subsurface lateral flow using Hydrus-2D and the calculated using water balance equation (Eq. 1) in the monocropping system and agroforestry system. R^2 indicates the determination coefficient, and RE indicates the relative error (Eqs. 3 and 4).

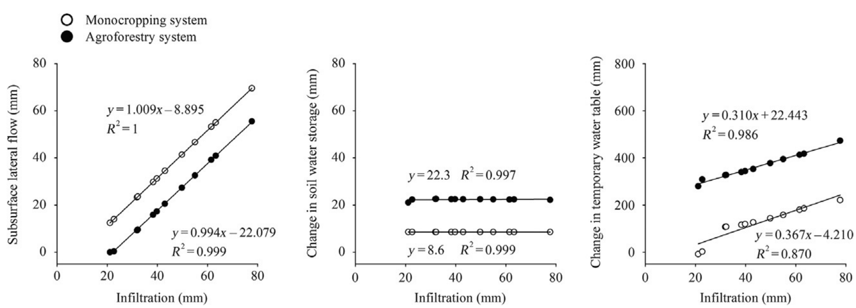

Fig. 8 Subsurface lateral flow, soil water storage change, and temporary water table change estimated under the virtual conditions of equal antecedent soil matric potential along the hillslope under different rainfall intensities by using Hydrus-2D modeling and their relations to infiltration water under different rainfall intensities for the monocropping system and agroforestry system.

temporary water table (Fig. 8). The predicted subsurface lateral flow and the change in temporary water table increased linearly with the increasing amount of infiltration water (rainfall minus runoff), while the predicted changes in soil water storage were constant under amount of different infiltration water. Comparing results between the two systems at the same infiltration, the AF treatment generated less subsurface lateral flow with greater soil water storage and a higher temporary water table elevation, compared with MC.

4. DISCUSSION

4.1 Generation of Subsurface Lateral Flow

The water budget estimated using Hydrus-2D code shows that the subsurface later flow was less in AF than in MC at all the comparable rainfall intensities (Table 2). This result is consistent with our previous study based on field monitoring data obtained over 3 years (Wang et al., 2011). Both studies demonstrate a larger domain and a longer resident time of water saturation in the deeper soil layers in AF than in MC during rainstorms. However, the previous study suggests that the reduced lateral flow was largely related to the increased evapotranspiration and tree canopy interception in AF (Wang et al., 2011). Here, we show that the reduced subsurface lateral flow was related to the increased tree canopy interception and soil water storage (by 12.4% and 6.4%, respectively, for the comparable 95.6 mm rainfall events) in AF compared to MC (Table 2). The increased canopy interception will lower infiltration water, while the increased soil water storage will increase the residence time of soil water in the soil profile, both of which will then reduce the subsurface lateral flow during rainstorms. The discrepancy on evapotranspiration and soil water storage is discussed later.

The evapotranspiration is higher in AF than in MC which will influence antecedent soil moisture before rainstorms and then may reduce subsurface lateral flow. The rainfall simulations have been carried out under conditions with negligible evapotranspiration and therefore are vital to test the hypothesis that subsurface lateral flow may change through modification of belowground processes by roots. With the simulation rainfall experiments, the effect of altered soil properties in AF on the reduction in subsurface lateral flow compared to MC was confirmed, though the effect was smaller than that of the tree canopy interception.

The rainfall stimulation experiments show negligible differences in the average overland flow values between AF and MC (Table 2), which is inconsistent with many other studies, which have reported that AF reduces the overland flow compared to MC (Udawatta et al., 2002; Wang et al., 2011). The different results of this study can be attributed to the high intensities of the simulated rainfalls, which may have caused saturation excess runoff as reported by Verbist et al. (2007). In addition, the plow pan found in MC may reduce vertical preferential flow and enhance the subsurface lateral flow above the plow pan compared with AF (Wang and Zhang, 2016).

When the plow layer is saturated in MC, overland flow may occur (Verbist et al., 2007). This will result in larger overland flow in MC than in AF even though infiltration can be increased after soil tillage which is included in the experimental conditions of this study. Therefore, the smaller overland flow in AF does not appear to be the reason for the reduced subsurface lateral flow compared to MC.

The spatial and temporal variations of the soil moisture during the rainfall events of 95.6 mm in this study show a large water saturation domain in deeper soil layers in both agroecosystems and disconnected water saturation domains between the top and deep soil layers in AF soon after the beginning of the rainfall (0.5 h) (Fig. 4), indicating vertical preferential flow in both agroecosystems, but likely with different pathways. The dye tracing experiments conducted before and after the rainfall in the same fields confirmed that the preferential flow through the plow pan below the plowed layer occurred mainly through well-connected cracks in MC and through relative disconnected large biopores developed by senile or decaying roots (Wang and Zhang, 2016). Field excavation from the studied fields showed that majority of peanut roots were distributed within the top 0.20 m soil, while majority of citrus roots were concentrated within the top 0.90 m soil (Table 1). Large of macropores from decaying roots are often reported in other ecosystems with vegetation dominated by trees (Schaik, 2010; Weiler and McDonnell, 2007; Zhang et al., 2015a,b). The existence of macropores is consistent with the disconnected water saturation domains in the soil profile of the AF system in this study. With the prolongation of the rainfall (0.5–2.0 h after the beginning of the rainfall), the water saturation domains became larger and connected along the slope and between the top and deep soil layers in MC and AF, respectively (Fig. 4). These results indicate that more soil water was retained in the soil profile in AF than in MC, resulting in the reduced subsurface lateral flow in AF. After the end of rainfall, the water saturation domain decreased in both systems, but was still larger in deep soil layers along the slope in AF than in MC, indicating slower drainage in AF. Therefore, the simulation rainfall experiments explicitly illustrate that the reduced subsurface lateral flow was attributed to the increased canopy interception and the increased water retention capacity in the soil profile due to different pathways of preferential vertical flow in AF compared to MC. The findings of this study and a dye tracing study (Wang and Zhang, 2016) provide useful insights to explore the mechanisms of subsurface lateral flow in Earth's CZ, which will benefit land use planners and land managers for supporting and sustaining terrestrial ecosystems (Banwart et al., 2012).

4.2 The Role of Soil Structure on Subsurface Lateral Flow Generation

The selected soil properties do not show significant differences between the two agroecosystems (Table 1), considering the short antecedent period for the differing agricultural practices and the small dimensions of the experimental plots. Therefore, the reduced subsurface lateral flow in AF could not be attributed to soil texture, but to altered soil structure and the resulting changes in hydraulic properties. Forests can retain more infiltrated water within soil profiles during rainfall (Bruijnzeel, 2004; Myers, 1983), as a part of the reservoir function of trees described in the literatures (Bruijnzeel, 2004; Myers, 1983). Mapa (1995) found higher soil water retention at any given water potential at all of the studied depths in reforested soils than in cultivated and grassland soils. For our research fields, the lab-determined soil water retention was higher for the 015–0.20 m soil depth in AF compared to MC and not different for the deeper soil layers between the two agroecosystems (Jing et al., 2008; Wang et al., 2013) (Fig. 9). However, at 0.2–0.9 m depth, the estimated van Genuchten parameter values using inverse modeling describe a water retention curve with lower water retention at lower soil matric potentials in AF than in MC. This is despite greater water retention in AF at near-saturation soil matric potentials (Fig. 9). These results indicate that the changed soil water retention curve cannot explain the increased water retention capacity and the reduced subsurface lateral flow in AF. The saturated hydraulic conductivity was greater at the 0.2–0.9 m depths compared to the top and deep soil layers in both agroecosystems (Table 2), suggesting presence of macropores at the 0.2–0.9 m depths. In AF, the subsurface later flow may be retarded due to the deeper root system, which may have not only intensified the structural connectivity through macropores in vertical direction, but also reduced the structural connectivity in the lateral direction. As if the subsurface lateral flow was retarded in AF, quick preferential flow would be quicker due to more macropores compared to MC. This would then generate a temporary water table over the impermeable layer generates (Fig. 4). Therefore, the increased water retention capacity in AF can be explained by the raised water table due to the presence of macropores and the reduced subsurface lateral flow.

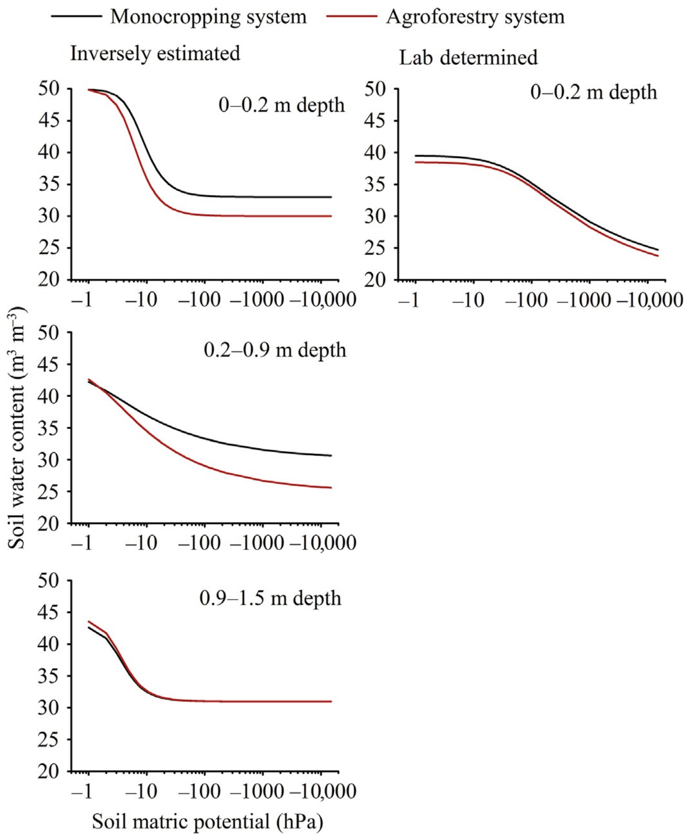

Fig. 9 Soil water retention curve at different soil depths in the monocropping system and agroforestry system, inversely estimated based on the van Genuchten parameters listed in Table 2 (*left*) and measured through pressure–membrane meter method using 100 cm^3 soil cores (Wang et al., 2013) (*right*).

There are few studies on the effect of land use on soil hydraulic properties in the lateral direction. The effects may be related to the development of soil micro- and mesopores due to the presence of fine roots or increased inputs of soil carbon as below-ground biomass and exudates, which promote the formation of large soil aggregates (Bogner et al., 2010; Mapa, 1995; Nair et al., 2010). Many studies have shown anisotropy of soil structure and hydraulic properties at landscape scale (Bottinelli et al., 2016; Jing et al., 2008; Soracco et al., 2010). The effects of macropores on the reduced subsurface lateral flow is indirectly reflected by the time delay of modeled peak flow rate compared to the observed values in different soil layers for both agroecosystems, but which is more profound in AF than in MC (Fig. 5).

The weaker goodness of fit in AF is likely due to the altered macropore system (Fig. 6), which is not represented by the altered soil water retention curve in the Hydrus-2D model.

To isolate the effects of soil structure change on the generation of subsurface lateral flow from the effect of antecedent soil moisture, a numerical experiment with Hydrus-2D was performed under the virtual condition of the same antecedent soil matric potential along the hillslope. The two agroecosystems showed a similar increasing trend with increasing effective rainfall intensity (the amount of infiltration water) in terms of the temporal water table and the subsurface lateral flow and no relation in terms of the soil water storage (Fig. 8). These results suggest that the effect of antecedent soil moisture or its related evapotranspiration can be ruled out as the reason for the reduced subsurface lateral flow in AF compared to MC. That is, the reduced subsurface lateral flow in AF is attributed to the increased tree canopy interception (or reduced infiltration) and the altered soil structure, which controls the anisotropy of soil hydraulic properties in vertical and lateral directions. Therefore, further studies are needed to elucidate the mechanisms how the tree root system shapes the anisotropy of soil hydraulic properties and consequently controls hillslope subsurface hydrological processes in Earth's CZ.

5. CONCLUSIONS

The present study coupled pedological and biological processes to explore the mechanisms through which agroforestry practices reduce subsurface lateral flow compared to monocropping system. The rainfall simulation experiments show lower infiltration water due to tree canopy interception, greater water retention capacity, and lower subsurface lateral flow in the agroforestry than in the monocropping system. The water partitioning and water movement is attributed to the deep roots of the agroforestry, which increase macropores in the vertical direction and alter micro- and mesopores in the lateral direction, resulting in more vertical preferential flow and less lateral flow within the soil profile and then enlarging soil water retention capacity. The numerical modeling at the virtual condition of the same antecedent soil moisture along the hillslope for all different rainfall intensities rules out the effect of antecedent soil moisture on the generation of subsurface lateral flow. Thus, although land use impacts evapotranspiration and then influences soil moisture conditions before rainfall, evapotranspiration may not be related to the effect of land use on generation of subsurface lateral flow.

This study suggests that the land use has strong effects on soil water mass and flow distribution due to the interaction between soils and roots. Therefore, understanding coupled pedological and biological processes in plant–soil–water systems are essential to understand the complexity of not only land surface but also subsoil ecological processes in Earth's CZ. Further studies need to elucidate the mechanisms by which these physical and biochemical processes connect to subsurface lateral flow processes, and also to quantify the magnitude of their relationship in Earth's CZ.

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